

Repsol Tensión Centauro

Naphthenic dielectric oil — Uninhibited

PRODUCT DATA SHEET

Description

TENSIÓN CENTAURO oils are light, naphthenic oils, particularly recommended for use as insulating fluid in electrical equipment such as power and distribution transformers, circuit breakers, rheostats, etc. In general, they can be used in any kind of electrical appliance requiring an oil bath which acts as a dielectric or coolant. Their production process ensures the lack of any solid, polar or water compounds, as well as the lack of any corrosive sulphur, which has been proven to be extremely damaging to transformers.

Classification and standards

TENSIÓN CENTAURO is an uninhibited oil that complies with the IEC 60296 and ASTM D3487 Type I standards.

Technical specifications

Values according to IEC 60296: 2012 standard.

Properties	Units	Method	Min.	Max.	Typical value
Physical properties					
Appearance		Visual		Bright and clear	
Density at 20 °C	g/cm ³	ISO 12185		0.895	0.87
Viscosity at 40 °C	mm ² /s	ASTM D445		12	9.6
Viscosity at 100 °C	mm ² /s	ASTM D445			2.4
Viscosity at -30 °C	mm ² /s	ASTM D445		1800	940
Pour point	°C	ASTM D5950		-45	-51
Interfacial tension	dynes/cm	ASTM D971	40		45
Chemical properties					
Acidity	mg KOH/g	ASTM D974		0.01	<0.01
Potentially corrosive sulphur		IEC 62535		Non-corrosive	
Corrosive sulphur		DIN 51353		Non-corrosive	
Corrosive sulphur		ASTM D1275B		Non-corrosive	
Antioxidant additive	% weight	IEC 60666		Exempt	
Water content	mg/kg	IEC 60814		30	15
Furfural content	mg/kg	IEC 61198		0.05	<0.05
Electrical properties					
Dielectric loss factor at 90 °C		IEC 60247		0.005	0.00075
Interfacial tension	mN/m	UNE 21320	40		46
Breakdown voltage, untreated	kV	IEC 60156	30		40
Breakdown voltage, treated	kV	IEC 60156	70		>70

Properties	Units	Method	Min.	Max.	Typical value
Oxidation stability — IEC 61125C (164 h / 120 °C)					
Total acidity	mg KOH/g	IEC 61125C		1.2	0.31
Sludge	% weight	IEC 61125C		0.8	0.097
Dielectric loss factor at 90 °C		IEC 61125C		0.5	0.055
Health, safety and the environment					
Flash point	°C	ASTM D93	135		150
PCA content	% weight	IP 346		3	<3
PCB content	% weight	IEC 61619		Exempt	

Note: Typical values correspond to current production. Whilst future production will conform to Repsol's specification, variations in these characteristics may occur. The guaranteed minimum/maximum values define the product specification.